

Part I

SUMMARY of Sigma Notation and Riemann Sums

Summary

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Useful formulas: $(1) \sum_{k=1}^n C = n \cdot C; \quad (2) \sum_{k=1}^n k = \frac{n(n+1)}{2}; \quad (3) \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad (3) \sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$

Useful example of application of rules for sums:

$$\sum_{k=1}^n (ak + b) = \sum_{k=1}^n ak + \sum_{k=1}^n b = a \sum_{k=1}^n k + \sum_{k=1}^n b = a \frac{n(n+1)}{2} + nb, \text{ where } a \text{ and } b \text{ constants.}$$

SUMMARY of Riemann Sums:

Riemann sum: $\sum_{k=1}^n f(x_k^*) \Delta x,$

Right Riemann sum: $x_k^* = x_k;$ Left Riemann sum: $x_k^* = x_{k-1};$ Midpoint Riemann sum: $x_k^* = \frac{x_{k-1} + x_k}{2}.$

Width of each of n rectangles on the interval $[a, b]: \quad \Delta x = \frac{b-a}{n}$

Grid points for interval $[a, b]:$

$$x_0 = a,$$

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$$x_{k-1} = a + (k-1)\Delta x = a + (k-1)\frac{b-a}{n},$$

$$x_k = a + k\Delta x = a + k\frac{b-a}{n},$$

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$$x_n = a + n\Delta x = a + n\frac{b-a}{n} = b$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 *Evaluate the sum.*

(a) $\sum_{k=1}^n 5$

(b) $\sum_{k=1}^{10} 5$

(c) $\sum_{k=1}^n 8k$

(d) $\sum_{k=1}^4 8k$

(e) $\sum_{k=1}^n (6k^2 - 8k - 5)$

(f) $\sum_{k=0}^2 \cos\left(\frac{\pi}{2}k\right)$

Problem 2

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Problem 2

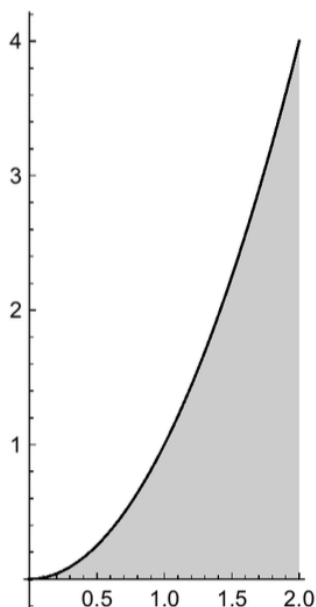
- Problem 2** (a) *If a function is positive and decreasing on an interval $[a, b]$, will a right Riemann sum underestimate or overestimate the area of the region under the graph of the function? Justify your answer.*
- (b) *If a function is positive and decreasing on an interval $[a, b]$, will a left Riemann sum underestimate or overestimate the area of the region under the graph of the function? Justify your answer.*

Problem 3

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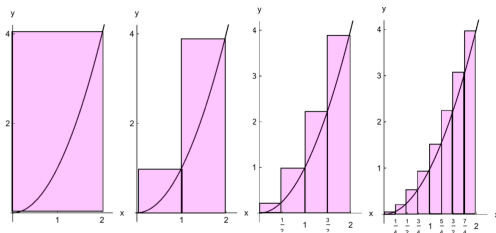
Problem 3

Problem 3 The graph of the function $f(x) = x^2$ is given in the figure.



Will a right Riemann sum approximation, for any value of n , be an underestimate or overestimate?

Approximate the shaded area using a right Riemann sum with $n = 1, 2, 4$, and 8 rectangles, as illustrated in the figure below.



- Approximate the shaded area using a right Riemann sum with $n = 1$ rectangles.
- Approximate the shaded area using a right Riemann sum with $n = 2$ rectangles.
- Approximate the shaded area using a right Riemann sum with $n = 4$ rectangles.

Problem 3

- (d) *Approximate the shaded area using a right Riemann sum with $n = 8$ rectangles.*

Problem 4

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Problem 4

Problem 4 A positive continuous function will have area approximated on the interval $[1, 6]$ using n rectangles.

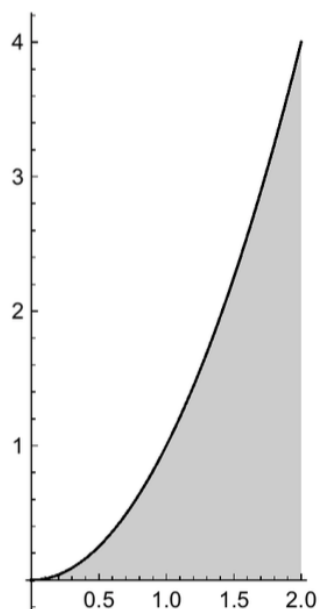
- (a) Find a formula for the grid point, x_k .
- (b) Find a formula for the sample point x_k^* if using a right Riemann sum.
- (c) Find a formula for the sample point x_k^* if using a left Riemann sum.
- (d) Find a formula for the sample point x_k^* if using a midpoint Riemann sum.

Problem 5

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Problem 5

Problem 5 The graph of the function $f(x) = x^2$ is given in the figure.



Find the exact value of the area of the shaded region. *HINT: Use a right Riemann sum with n rectangles, and then take the limit as $n \rightarrow \infty$.*

Problem 6

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Problem 6

Problem 6 Consider a Riemann sum with n rectangles for the function f on the interval $[a, b]$.

Use geometry to find the limit of Riemann sums as $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x \text{ ?}$$

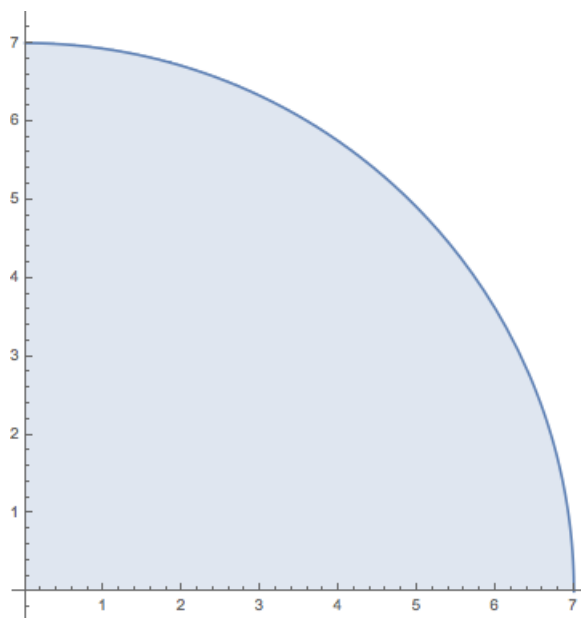
(a) $f(x) = x, [0, 3];$

(b) $f(x) = |x|, [-3, 3];$

Problem 7

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Problem 7 A part of a circle is shown in the figure.



- (a) If we express the area of the shaded region in the figure as the limit of Riemann sums

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x,$$

find the function f and the interval $[a, b]$.

- (b) Compute the limit.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{49 - (x_k^*)^2} \cdot \frac{7}{n}$$

Problem 8

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Problem 8

Problem 8 We want to approximate the area under the curve using a right Riemann Sum with the given value of n . Write the sum in summation notation and evaluate it.

(a) $\sin(x)$, $\left[0, \frac{\pi}{2}\right]$, $n = 3$

(b) $f(x) = x^2 - 9x + 18$, $[7, 10]$, $n = 6$